

ECOLEARN OPERATING PLAN

Web + mobile costs for 5,000-10,000 users

A scale-as-needed budget for a Delaware school pilot and a larger DSWA-backed rollout. The figures separate recurring platform costs, usage-sensitive AI, store fees, and the professional work needed for a safe education product.

\$108-\$419	\$149-\$699	1 call	\$124
5,000-user monthly platform range	10,000-user monthly platform range	LLM requests per visual scan	store accounts: first-year minimum

BOTTOM LINE

The existing free stack is suitable for development and a small supervised pilot. For 5,000-10,000 users, fund Supabase Pro, production email, Netlify headroom, a commercial map provider, OpenRouter credits with hard caps, monitoring, and an Expo plan matched to mobile update MAU.

Planning assumptions

These are capacity-planning scenarios, not a claim that every registered user is active at the same moment.

Input	5,000-user case	10,000-user case
Registered users	5,000	10,000
Monthly active users	About 2,500	5,000-7,500
Visual scans/month	5,000	15,000
Transactional email/month	Up to 10,000	About 20,000
Traffic shape	School-day bursts	Larger classroom bursts
Image retention	No scan-photo storage	No scan-photo storage
Concurrency	Measured, not 5,000 simultaneous	Load test classroom peaks

What the current PDF gets right

The attached two-page plan is visually polished and correctly identifies the Apple, Google Play, OpenRouter, Supabase Pro, custom-domain, and ownership decisions needed to launch. Keep it as the short executive brief.

What it was missing

It did not model actual 5,000/10,000-user usage, Netlify hosting credits, production SMTP, monitoring, commercial maps, AI scan volume, backups/restore testing, school privacy and accessibility work, or the difference between cloud bills and paid maintenance labor. This report fills those gaps.

Recommended architecture at this size

- Netlify serves the Vite SPA; Supabase owns Auth, Postgres, RPCs, Edge Functions, and the mirrored DNREC catalog.
- One compressed photo produces one OpenRouter vision call. Deterministic server-side catalog matching validates the model hint; do not add one LLM call per detected object.
- DNREC remains the authority for item protocols and official mapped solutions. Cache catalog data and alert when its schema or sync changes.
- Expo/EAS produces iOS and Android builds and optional over-the-air updates. Store accounts are separate from Expo fees.

Recurring service budget

Service	5,000 users	10,000 users	Why / trigger
Supabase	\$25	\$25-\$85	Pro includes 100k MAU, 8 GB disk, 250 GB egress, and 7-day backups. Add compute only after measured latency or connection pressure.
Netlify	\$9-\$20	\$20-\$40	Personal includes 1,000 credits; Pro starts with 3,000. Keep auto-recharge off until an owner approves the budget.
Resend SMTP	\$20	\$20	Pro includes 50,000 transactional emails and removes the free daily cap.
OpenRouter	\$10-\$100	\$30-\$300	Usage-sensitive. Replace the range with actual cost per successful scan and a 30 percent safety factor.
MapTiler	\$25	\$25	Flex supports commercial use, 25,000 map sessions, and 500,000 API requests.
Expo/EAS	\$19-\$199	\$19-\$199	Starter has 3,000 update MAU; Production has 50,000. Use overage calculator before jumping tiers.
Monitoring	\$0-\$30	\$10-\$30	Start with uptime, cron/sync heartbeat, JS errors, function failures, and an alert owner.
TOTAL	\$108-\$419	\$149-\$699	Monthly total. Excludes labor, tax, domain, store accounts, reviews, and unusually high AI usage.

AI cost guardrail

Monthly AI budget = successful visual requests x measured 30-day average cost/request x 1.30 safety factor. Use a production-only OpenRouter key, a hard monthly cap, per-user hourly limits, a global atomic reservation, no automatic retry loop, and a free text/catalog fallback when the cap is reached.

Example cost per scan	5,000 scans	15,000 scans	+30% reserve
\$0.002	\$10	\$30	\$13 / \$39
\$0.010	\$50	\$150	\$65 / \$195
\$0.020	\$100	\$300	\$130 / \$390

One-time, annual, and people costs

Item	Planning amount	Notes
Apple Developer Program	\$99/year	Eligible nonprofits, accredited education institutions, and government entities can request a fee waiver.
Google Play full distribution	\$25 one time	Required for public Play distribution. A limited no-fee option is restricted to 20 devices.
Domain renewal	\$15-\$40/year	Move domain, DNS, MFA, recovery contacts, and auto-renewal to the sponsoring organization.
Privacy/legal review	Quote separately	Under-13 classroom deployment, consent, retention, deletion, contracts, incident response, and vendor terms.
Accessibility review	Quote separately	WCAG 2.2 AA, captions/transcripts, keyboard/screen-reader QA, and iOS/Android accessibility testing.
Security review	Quote separately	Threat model, RLS/function review, secrets, dependency posture, backups, and incident exercise.
Engineering/content support	Staff or retainer	Updates, release QA, DNREC/DSWA content maintenance, store reviews, and user support.

Do not buy everything on day one

Start with Supabase Pro, production SMTP, Netlify Personal/Pro, \$25-\$100 of OpenRouter credits, and free monitoring. Add MapTiler when nearby search is promoted and move Expo from Starter to Production only when mobile update MAU or release operations justify it.

Readiness gates before inviting schools

- Agree on the teacher/student identity model and obtain the required DSWA, school, parent, privacy, and legal decisions.
- Run authenticated web/mobile regression tests and a classroom-burst load test against staging.
- Verify backup restore, account deletion, rate limits, spend caps, DNREC sync alerts, and incident ownership.
- Measure scanner success, no-match rate, per-scan cost, p95 latency, and category confusion with a Delaware validation set.
- Provide captions or accessible alternatives for DSWA videos and a fallback when school filters block YouTube.

Decision summary

Stage	Fund now	Wait for evidence
Development	Small OpenRouter credit balance; preserve hard app limits	Store accounts, commercial maps, Expo Production
Supervised pilot	Supabase Pro, SMTP, Netlify headroom, monitoring, Expo Starter	Larger compute, enterprise plans
5,000 users	MapTiler Flex, measured AI reserve, restore drill, named incident owner	Expo Production unless update MAU/ops need it
10,000 users	Load-tested compute, Pro hosting, monitoring, privacy/accessibility/security review	Enterprise contracts unless DSWA requires SLA/SSO

Official sources checked August 20, 2026

- Supabase pricing
- Netlify pricing and usage credits
- Expo Application Services pricing
- Resend pricing
- OpenRouter pricing
- MapTiler Cloud pricing
- Apple Developer Program enrollment
- Google Play Console registration
- Better Stack monitoring pricing

Model limitations

Vendor prices and allowances can change. The AI range is intentionally scenario-based because the configured vision model and provider routing determine image/token cost. Reprice before approval and replace assumptions with telemetry after the first 30 days of a funded pilot.